



MATH155: Analytic Trigonometry

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Learning Objectives

- Apply the Law of Sines to find missing sides and angles
- Apply the Law of Cosines to find missing sides and angles
- Solve triangles using the appropriate law

Simplify each expression completely. Show all steps and circle your final answer.

Law of Cosines

1. In triangle ABC, $a = 4$, $b = 3$, and angle $C = 60^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 4, b = 3, C = 60^\circ$$

Answer: _____

2. In triangle ABC, $a = 4$, $b = 9$, and angle $C = 30^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 4, b = 9, C = 30^\circ$$

Answer: _____

3. In triangle ABC, $a = 10$, $b = 7$, and angle $C = 60^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 10, b = 7, C = 60^\circ$$

Answer: _____

4. In triangle ABC, $a = 5$, $b = 10$, and angle $C = 120^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 5, b = 10, C = 120^\circ$$

Answer: _____

5. In triangle ABC, $a = 6$, $b = 6$, and angle $C = 45^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 6, b = 6, C = 45^\circ$$

Answer: _____



6. In triangle ABC, $a = 5$, $b = 4$, and angle $C = 45^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 5, b = 4, C = 45^\circ$$

Answer: _____

7. In triangle ABC, $a = 4$, $b = 7$, and angle $C = 90^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 4, b = 7, C = 90^\circ$$

Answer: _____

8. In triangle ABC, $a = 4$, $b = 7$, and angle $C = 90^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 4, b = 7, C = 90^\circ$$

Answer: _____

Law of Cosines — find missing angle

9. In triangle ABC, $a = 4$, $b = 6$, $c = 9$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 4, b = 6, c = 9$$

Answer: _____

10. In triangle ABC, $a = 6$, $b = 4$, $c = 7$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 6, b = 4, c = 7$$

Answer: _____

11. In triangle ABC, $a = 5$, $b = 8$, $c = 8$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 5, b = 8, c = 8$$

Answer: _____

12. In triangle ABC, $a = 4$, $b = 5$, $c = 5$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 4, b = 5, c = 5$$

Answer: _____



13. In triangle ABC, $a = 5$, $b = 4$, $c = 9$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 5, b = 4, c = 9$$

Answer: _____

14. In triangle ABC, $a = 7$, $b = 7$, $c = 6$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 7, b = 7, c = 6$$

Answer: _____

15. In triangle ABC, $a = 4$, $b = 4$, $c = 7$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 4, b = 4, c = 7$$

Answer: _____

16. In triangle ABC, $a = 6$, $b = 8$, $c = 8$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 6, b = 8, c = 8$$

Answer: _____

Law of Sines

17. In triangle ABC, angle $A = 45^\circ$, angle $B = 45^\circ$, and side $a = 6$. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 45^\circ, B = 45^\circ, a = 6$$

Answer: _____

18. In triangle ABC, angle $A = 60^\circ$, angle $B = 75^\circ$, and side $a = 6$. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 60^\circ, B = 75^\circ, a = 6$$

Answer: _____

19. In triangle ABC, angle $A = 45^\circ$, angle $B = 45^\circ$, and side $a = 12$. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 45^\circ, B = 45^\circ, a = 12$$

Answer: _____



20. In triangle ABC, angle A = 60° , angle B = 45° , and side a = 11. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 60^\circ, B = 45^\circ, a = 11$$

Answer: _____

21. In triangle ABC, angle A = 45° , angle B = 75° , and side a = 6. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 45^\circ, B = 75^\circ, a = 6$$

Answer: _____

22. In triangle ABC, angle A = 45° , angle B = 45° , and side a = 10. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 45^\circ, B = 45^\circ, a = 10$$

Answer: _____

23. In triangle ABC, angle A = 60° , angle B = 90° , and side a = 12. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 60^\circ, B = 90^\circ, a = 12$$

Answer: _____

Law of Sines — find missing angle

24. In triangle ABC, angle A = 45° , side a = 11, and side b = 10. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 45^\circ, a = 11, b = 10$$

Answer: _____

25. In triangle ABC, angle A = 30° , side a = 10, and side b = 9. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 30^\circ, a = 10, b = 9$$

Answer: _____

26. In triangle ABC, angle A = 60° , side a = 8, and side b = 10. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 60^\circ, a = 8, b = 10$$

Answer: _____



27. In triangle ABC, angle $A = 30^\circ$, side $a = 10$, and side $b = 3$. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 30^\circ, a = 10, b = 3$$

Answer: _____

28. In triangle ABC, angle $A = 45^\circ$, side $a = 7$, and side $b = 6$. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 45^\circ, a = 7, b = 6$$

Answer: _____

29. In triangle ABC, angle $A = 45^\circ$, side $a = 9$, and side $b = 7$. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 45^\circ, a = 9, b = 7$$

Answer: _____

30. In triangle ABC, angle $A = 60^\circ$, side $a = 5$, and side $b = 4$. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 60^\circ, a = 5, b = 4$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Law of Cosines — find missing angle, Law of Sines, Law of Sines — find missing angle, Law of Cosines. All answers verified by independent computation.

Solutions



Law of Cosines

1. In triangle ABC, $a = 4$, $b = 3$, and angle $C = 60^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 4, b = 3, C = 60^\circ$$

$$\rightarrow \text{Law of Cosines: } c^2 = a^2 + b^2 - 2ab \cos(C).$$

$$\rightarrow c^2 = 4^2 + 3^2 - 2(4)(3) \cos(60^\circ).$$

$$\rightarrow c^2 = 16 + 9 - 24 \cdot 0.5 = 13.0.$$

$$\rightarrow c = \sqrt{13.0} = 3.61.$$

Answer: $c = 3.61$

2. In triangle ABC, $a = 4$, $b = 9$, and angle $C = 30^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 4, b = 9, C = 30^\circ$$

$$\rightarrow \text{Law of Cosines: } c^2 = a^2 + b^2 - 2ab \cos(C).$$

$$\rightarrow c^2 = 4^2 + 9^2 - 2(4)(9) \cos(30^\circ).$$

$$\rightarrow c^2 = 16 + 81 - 72 \cdot 0.866 = 34.6462.$$

$$\rightarrow c = \sqrt{34.6462} = 5.89.$$

Answer: $c = 5.89$

3. In triangle ABC, $a = 10$, $b = 7$, and angle $C = 60^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 10, b = 7, C = 60^\circ$$

$$\rightarrow \text{Law of Cosines: } c^2 = a^2 + b^2 - 2ab \cos(C).$$

$$\rightarrow c^2 = 10^2 + 7^2 - 2(10)(7) \cos(60^\circ).$$

$$\rightarrow c^2 = 100 + 49 - 140 \cdot 0.5 = 79.0.$$

$$\rightarrow c = \sqrt{79.0} = 8.89.$$

Answer: $c = 8.89$

4. In triangle ABC, $a = 5$, $b = 10$, and angle $C = 120^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 5, b = 10, C = 120^\circ$$

$$\rightarrow \text{Law of Cosines: } c^2 = a^2 + b^2 - 2ab \cos(C).$$

$$\rightarrow c^2 = 5^2 + 10^2 - 2(5)(10) \cos(120^\circ).$$

$$\rightarrow c^2 = 25 + 100 - 100 \cdot (-0.5) = 175.0.$$

$$\rightarrow c = \sqrt{175.0} = 13.23.$$

Answer: $c = 13.23$



5. In triangle ABC, $a = 6$, $b = 6$, and angle $C = 45^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 6, b = 6, C = 45^\circ$$

$$\rightarrow \text{Law of Cosines: } c^2 = a^2 + b^2 - 2ab \cos(C).$$

$$\rightarrow c^2 = 6^2 + 6^2 - 2(6)(6) \cos(45^\circ).$$

$$\rightarrow c^2 = 36 + 36 - 72 \cdot 0.7071 = 21.0883.$$

$$\rightarrow c = \sqrt{21.0883} = 4.59.$$

Answer: $c = 4.59$

6. In triangle ABC, $a = 5$, $b = 4$, and angle $C = 45^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 5, b = 4, C = 45^\circ$$

$$\rightarrow \text{Law of Cosines: } c^2 = a^2 + b^2 - 2ab \cos(C).$$

$$\rightarrow c^2 = 5^2 + 4^2 - 2(5)(4) \cos(45^\circ).$$

$$\rightarrow c^2 = 25 + 16 - 40 \cdot 0.7071 = 12.7157.$$

$$\rightarrow c = \sqrt{12.7157} = 3.57.$$

Answer: $c = 3.57$

7. In triangle ABC, $a = 4$, $b = 7$, and angle $C = 90^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 4, b = 7, C = 90^\circ$$

$$\rightarrow \text{Law of Cosines: } c^2 = a^2 + b^2 - 2ab \cos(C).$$

$$\rightarrow c^2 = 4^2 + 7^2 - 2(4)(7) \cos(90^\circ).$$

$$\rightarrow c^2 = 16 + 49 - 56 \cdot 0.0 = 65.0.$$

$$\rightarrow c = \sqrt{65.0} = 8.06.$$

Answer: $c = 8.06$

8. In triangle ABC, $a = 4$, $b = 7$, and angle $C = 90^\circ$. Use the Law of Cosines to find side c (round to 2 decimal places).

$$a = 4, b = 7, C = 90^\circ$$

$$\rightarrow \text{Law of Cosines: } c^2 = a^2 + b^2 - 2ab \cos(C).$$

$$\rightarrow c^2 = 4^2 + 7^2 - 2(4)(7) \cos(90^\circ).$$

$$\rightarrow c^2 = 16 + 49 - 56 \cdot 0.0 = 65.0.$$

$$\rightarrow c = \sqrt{65.0} = 8.06.$$

Answer: $c = 8.06$



Law of Cosines — find missing angle

9. In triangle ABC, $a = 4$, $b = 6$, $c = 9$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 4, b = 6, c = 9$$

→ Rearrange the Law of Cosines for angle C:

$$\rightarrow \cos(C) = (a^2 + b^2 - c^2) / (2ab) = (4^2 + 6^2 - 9^2) / (2 \cdot 4 \cdot 6).$$

$$\rightarrow \cos(C) = (16 + 36 - 81) / 48 = -0.6042.$$

$$\rightarrow C = \arccos(-0.6042) = 127.2^\circ.$$

Answer: $C = 127.2^\circ$

10. In triangle ABC, $a = 6$, $b = 4$, $c = 7$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 6, b = 4, c = 7$$

→ Rearrange the Law of Cosines for angle C:

$$\rightarrow \cos(C) = (a^2 + b^2 - c^2) / (2ab) = (6^2 + 4^2 - 7^2) / (2 \cdot 6 \cdot 4).$$

$$\rightarrow \cos(C) = (36 + 16 - 49) / 48 = 0.0625.$$

$$\rightarrow C = \arccos(0.0625) = 86.4^\circ.$$

Answer: $C = 86.4^\circ$

11. In triangle ABC, $a = 5$, $b = 8$, $c = 8$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 5, b = 8, c = 8$$

→ Rearrange the Law of Cosines for angle C:

$$\rightarrow \cos(C) = (a^2 + b^2 - c^2) / (2ab) = (5^2 + 8^2 - 8^2) / (2 \cdot 5 \cdot 8).$$

$$\rightarrow \cos(C) = (25 + 64 - 64) / 80 = 0.3125.$$

$$\rightarrow C = \arccos(0.3125) = 71.8^\circ.$$

Answer: $C = 71.8^\circ$

12. In triangle ABC, $a = 4$, $b = 5$, $c = 5$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 4, b = 5, c = 5$$

→ Rearrange the Law of Cosines for angle C:

$$\rightarrow \cos(C) = (a^2 + b^2 - c^2) / (2ab) = (4^2 + 5^2 - 5^2) / (2 \cdot 4 \cdot 5).$$

$$\rightarrow \cos(C) = (16 + 25 - 25) / 40 = 0.4.$$

$$\rightarrow C = \arccos(0.4) = 66.4^\circ.$$

Answer: $C = 66.4^\circ$



13. In triangle ABC, $a = 5$, $b = 4$, $c = 9$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 5, b = 4, c = 9$$

→ Rearrange the Law of Cosines for angle C:

$$\rightarrow \cos(C) = (a^2 + b^2 - c^2) / (2ab) = (5^2 + 4^2 - 9^2) / (2 \cdot 5 \cdot 4).$$

$$\rightarrow \cos(C) = (25 + 16 - 81) / 40 = -1.0.$$

$$\rightarrow C = \arccos(-1.0) = 180.0^\circ.$$

Answer: $C = 180.0^\circ$

14. In triangle ABC, $a = 7$, $b = 7$, $c = 6$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 7, b = 7, c = 6$$

→ Rearrange the Law of Cosines for angle C:

$$\rightarrow \cos(C) = (a^2 + b^2 - c^2) / (2ab) = (7^2 + 7^2 - 6^2) / (2 \cdot 7 \cdot 7).$$

$$\rightarrow \cos(C) = (49 + 49 - 36) / 98 = 0.6327.$$

$$\rightarrow C = \arccos(0.6327) = 50.8^\circ.$$

Answer: $C = 50.8^\circ$

15. In triangle ABC, $a = 4$, $b = 4$, $c = 7$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 4, b = 4, c = 7$$

→ Rearrange the Law of Cosines for angle C:

$$\rightarrow \cos(C) = (a^2 + b^2 - c^2) / (2ab) = (4^2 + 4^2 - 7^2) / (2 \cdot 4 \cdot 4).$$

$$\rightarrow \cos(C) = (16 + 16 - 49) / 32 = -0.5312.$$

$$\rightarrow C = \arccos(-0.5312) = 122.1^\circ.$$

Answer: $C = 122.1^\circ$

16. In triangle ABC, $a = 6$, $b = 8$, $c = 8$. Use the Law of Cosines to find angle C in degrees (round to 1 decimal place).

$$a = 6, b = 8, c = 8$$

→ Rearrange the Law of Cosines for angle C:

$$\rightarrow \cos(C) = (a^2 + b^2 - c^2) / (2ab) = (6^2 + 8^2 - 8^2) / (2 \cdot 6 \cdot 8).$$

$$\rightarrow \cos(C) = (36 + 64 - 64) / 96 = 0.375.$$

$$\rightarrow C = \arccos(0.375) = 68.0^\circ.$$

Answer: $C = 68.0^\circ$



Law of Sines

17. In triangle ABC, angle A = 45°, angle B = 45°, and side a = 6. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 45^\circ, B = 45^\circ, a = 6$$

$$\rightarrow \text{Law of Sines: } a / \sin(A) = b / \sin(B).$$

$$\rightarrow b = a * \sin(B) / \sin(A) = 6 * \sin(45^\circ) / \sin(45^\circ).$$

$$\rightarrow b = 6 * 0.7071 / 0.7071 = 6.0.$$

Answer: $b = 6.0$

18. In triangle ABC, angle A = 60°, angle B = 75°, and side a = 6. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 60^\circ, B = 75^\circ, a = 6$$

$$\rightarrow \text{Law of Sines: } a / \sin(A) = b / \sin(B).$$

$$\rightarrow b = a * \sin(B) / \sin(A) = 6 * \sin(75^\circ) / \sin(60^\circ).$$

$$\rightarrow b = 6 * 0.9659 / 0.866 = 6.69.$$

Answer: $b = 6.69$

19. In triangle ABC, angle A = 45°, angle B = 45°, and side a = 12. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 45^\circ, B = 45^\circ, a = 12$$

$$\rightarrow \text{Law of Sines: } a / \sin(A) = b / \sin(B).$$

$$\rightarrow b = a * \sin(B) / \sin(A) = 12 * \sin(45^\circ) / \sin(45^\circ).$$

$$\rightarrow b = 12 * 0.7071 / 0.7071 = 12.0.$$

Answer: $b = 12.0$

20. In triangle ABC, angle A = 60°, angle B = 45°, and side a = 11. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 60^\circ, B = 45^\circ, a = 11$$

$$\rightarrow \text{Law of Sines: } a / \sin(A) = b / \sin(B).$$

$$\rightarrow b = a * \sin(B) / \sin(A) = 11 * \sin(45^\circ) / \sin(60^\circ).$$

$$\rightarrow b = 11 * 0.7071 / 0.866 = 8.98.$$

Answer: $b = 8.98$

21. In triangle ABC, angle A = 45°, angle B = 75°, and side a = 6. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 45^\circ, B = 75^\circ, a = 6$$

$$\rightarrow \text{Law of Sines: } a / \sin(A) = b / \sin(B).$$

$$\rightarrow b = a * \sin(B) / \sin(A) = 6 * \sin(75^\circ) / \sin(45^\circ).$$

$$\rightarrow b = 6 * 0.9659 / 0.7071 = 8.2.$$

Answer: $b = 8.2$



22. In triangle ABC, angle A = 45°, angle B = 45°, and side a = 10. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 45^\circ, B = 45^\circ, a = 10$$

→ Law of Sines: $a / \sin(A) = b / \sin(B)$.

$$\rightarrow b = a * \sin(B) / \sin(A) = 10 * \sin(45^\circ) / \sin(45^\circ).$$

$$\rightarrow b = 10 * 0.7071 / 0.7071 = 10.0.$$

Answer: $b = 10.0$

23. In triangle ABC, angle A = 60°, angle B = 90°, and side a = 12. Use the Law of Sines to find side b (round to 2 decimal places).

$$A = 60^\circ, B = 90^\circ, a = 12$$

→ Law of Sines: $a / \sin(A) = b / \sin(B)$.

$$\rightarrow b = a * \sin(B) / \sin(A) = 12 * \sin(90^\circ) / \sin(60^\circ).$$

$$\rightarrow b = 12 * 1.0 / 0.866 = 13.86.$$

Answer: $b = 13.86$



Law of Sines — find missing angle

24. In triangle ABC, angle A = 45° , side a = 11, and side b = 10. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 45^\circ, a = 11, b = 10$$

$$\rightarrow \text{Law of Sines: } \sin(A)/a = \sin(B)/b.$$

$$\rightarrow \sin(B) = b \cdot \sin(A)/a = 10 \cdot \sin(45^\circ)/11 = 0.6428.$$

$$\rightarrow B = \arcsin(0.6428) = 40.0^\circ.$$

Answer: $B = 40.0^\circ$

25. In triangle ABC, angle A = 30° , side a = 10, and side b = 9. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 30^\circ, a = 10, b = 9$$

$$\rightarrow \text{Law of Sines: } \sin(A)/a = \sin(B)/b.$$

$$\rightarrow \sin(B) = b \cdot \sin(A)/a = 9 \cdot \sin(30^\circ)/10 = 0.45.$$

$$\rightarrow B = \arcsin(0.45) = 26.7^\circ.$$

Answer: $B = 26.7^\circ$

26. In triangle ABC, angle A = 60° , side a = 8, and side b = 10. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 60^\circ, a = 8, b = 10$$

$$\rightarrow \text{Law of Sines: } \sin(A)/a = \sin(B)/b.$$

$$\rightarrow \sin(B) = b \cdot \sin(A)/a = 10 \cdot \sin(60^\circ)/8 = 1.0.$$

$$\rightarrow B = \arcsin(1.0) = 90.0^\circ.$$

Answer: $B = 90.0^\circ$

27. In triangle ABC, angle A = 30° , side a = 10, and side b = 3. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 30^\circ, a = 10, b = 3$$

$$\rightarrow \text{Law of Sines: } \sin(A)/a = \sin(B)/b.$$

$$\rightarrow \sin(B) = b \cdot \sin(A)/a = 3 \cdot \sin(30^\circ)/10 = 0.15.$$

$$\rightarrow B = \arcsin(0.15) = 8.6^\circ.$$

Answer: $B = 8.6^\circ$

28. In triangle ABC, angle A = 45° , side a = 7, and side b = 6. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 45^\circ, a = 7, b = 6$$

$$\rightarrow \text{Law of Sines: } \sin(A)/a = \sin(B)/b.$$

$$\rightarrow \sin(B) = b \cdot \sin(A)/a = 6 \cdot \sin(45^\circ)/7 = 0.6061.$$

$$\rightarrow B = \arcsin(0.6061) = 37.3^\circ.$$

Answer: $B = 37.3^\circ$



29. In triangle ABC, angle $A = 45^\circ$, side $a = 9$, and side $b = 7$. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 45^\circ, a = 9, b = 7$$

$$\rightarrow \text{Law of Sines: } \sin(A)/a = \sin(B)/b.$$

$$\rightarrow \sin(B) = b \cdot \sin(A)/a = 7 \cdot \sin(45^\circ)/9 = 0.55.$$

$$\rightarrow B = \arcsin(0.55) = 33.4^\circ.$$

Answer: $B = 33.4^\circ$

30. In triangle ABC, angle $A = 60^\circ$, side $a = 5$, and side $b = 4$. Use the Law of Sines to find angle B in degrees (round to 1 decimal place).

$$A = 60^\circ, a = 5, b = 4$$

$$\rightarrow \text{Law of Sines: } \sin(A)/a = \sin(B)/b.$$

$$\rightarrow \sin(B) = b \cdot \sin(A)/a = 4 \cdot \sin(60^\circ)/5 = 0.6928.$$

$$\rightarrow B = \arcsin(0.6928) = 43.9^\circ.$$

Answer: $B = 43.9^\circ$

